Project Title



Capstone Portfolio Template 2026-2027

Team Code: 16

IATS: El Sewedy

Grade: Junior

Team Members:

Name	St. Code
1- Anas Samir Ahmed (Leader)	526008
2- mohamed khaled	526040
3- malak mahammed saied mahammed	526090
4- mariam mohamed ramadan	526086
5- Hamza Ahmed Mohamed Sabr	526015

Supervised by:

Eng.

Eng.

ABSTRACT

Mashwary is a smart ride-booking website prototype that combines ease of use, safety, and

privacy. Users choose their pickup point and destination and follow the trip live on the map.

They can also share their live location with a trusted person. The platform includes a safety

camera system inside drivers' cars, designed to protect the privacy of both passengers and

drivers. Passengers can choose a female driver when one is available. The website supports

Arabic and English.

Introduction

Okay so here's the thing. The whole problem comes down to trust. People don't feel getting into a car with a stranger.. Honestly it’s even worse if you're a girl or a woman. The riders are students, employees people just going out.. They’re scared of unknown drivers. They’re scared of riding at night. They just want someone to be tracking them the way. They want to relax, not stress.

The apps that exist right now? They’re a pain in Arabic. There's no way to pick a driver. The privacy stuff is vague.. Family can’t easily follow the trip.

So that’s where we started. We did the survey, the affinity map, the empathy map. And grouped everything into five themes:

• **Safety**. People are scared of drivers. They want cameras in the car. They want to share the trip with someone they trust.
• **Privacy**. People worry about where the camera footage goes. Drivers want their privacy too. There should be rules, on who’s allowed to watch recordings. • **. Trust**. A lot of women prefer a driver. So give them the option when there’s one. Let people see driver info before they get in.
• **Ease of Use**. Set pickup and destination. Track the trip live on the map. Keep booking to a few steps.
• **Language**. English both. Easy to switch. Words that actually make sense to peopl

What is the challenge or problem your project will solve?

So the real problem we're trying to fix is this. getting around every day is stressful and very unsafe, and it's worse if you're a girl or a woman. the usual ways people do it, like flagging down some random car or word of mouth or apps that don't really fit, just don't cut it. you don't know who's driving you, nobody's actually tracking your trip, the privacy rules are all vague (even getting kidnapped is a whole new problem, we’re looking forward to investigate), and Arabic support feels like an afterthought. so we're building one centralized platform for booking rides that fixes all that. safety cameras in the car, live location sharing with someone you trust, the option to pick a female driver, and full Arabic and English support. basically we want every trip to feel safe, private, and easy, no stress. Its made for customer service rather than just money, Good loyal customers > Money overfilling our banks but the reviews are bad or people trying to accuse us of unsafety and vague measurements and decisions.

What are the roles of each team member, and how will these roles be rotated?

Our team comprises of the following members:

- Malak Mahommad **:** Researcher
- Hamza Ahmed :Designer
    - Responsible for conducting market research, identifying user needs, and

creating the user interface and user experience (UI/UX) design.

- **Anas Samir Ahmed:** Developer
  - Responsible for the development and implementation of the mobile application, including backend and frontend development.
- Mariam Mohammad**:** Tester
  - Responsible for testing the application's functionality, usability, and performance to ensure a seamless user experience.

To foster collaboration and skill development, we will periodically rotate roles among team members. This rotation will allow each member to gain a comprehensive understanding of the project and contribute to different aspects of the development process.

Table of Contents

Section I: The problem background. 4 1.1 Sustainable Development Goals (SDGs) addressed 5 1.2 Empathize 6

Section II: Ideas and solutions 7 2.1 Define 8 2.2 Task Review: 8 2.3 Problem Statement: 8 2.4 Prior Solutions: 8 2.5 Ideate: 10 2.6 Problem Solution: 10 2.7 Our Selected Solution: 10 2.8 Solution Advantages & Disadvantages: 10

Section III: Experiment ideas and solutions 12 3.1 Task Review: 13 3.2 Methods used in Requirement Gathering: 13 3.3 Use Case Diagram: 13 3.4 Activity Diagram: 14 3.5 Sketches: 15 3.6 Linkable Prototype 15 3.7 Test and Evaluate: 16 3.8 Refine the design. 19

Section IV: Analyzing and communicating 20 4.1 Recommendations 21 4.2 Learning Outcomes 22

Competency Related: 22 4.3 Communication 23 4.4 References 24 Section V: Business Model Canvas Template 25

Section I: The problem background.

1.1 Sustainable Development Goals (SDGs) addressed



WE as mashwary are looking forward to support Gender Equality, Quality Eduction (Through Female Training Center [FTC]), Decent Work and Economic Growth by giving out jobs and training applications to females who like driving and cars overall

In this chapter, we will discuss Sustainable Development Goals that our project supports and how this project can help in sustainable development.



•SDG's addressed:

- Goal 8: Decent Work and Economic Growth
- Goal 11: Sustainable Cities and Communities

1.2 Empathize

- First is the people we are going to support and designed our application/idea for

We are going to support all kinds of people including females *specially*, We are designing our application toward females specially, Customers themselves have their own speciality like full filters for picking their drivers and who they wanna talk to and WHEN, allowing for scheduled tasks, drives, trips, etc

- Second is the needs from users and challenges we're going to face

Problems are listed within our boundaries and limitations;

- **Driving safety includes Females not trusting MEN so we're going to improve our system to make THEM only be with themselves (e.g ; Females with females, Males with males)**
- **No long trips or too expensive trips, We're going to fix it by using the Uber Account system, the more people use our app, the more coupons they will get for each hour they will be using the app, AFK'ing is not allowed because they must actually use it by playing built in games to grind for coupons that include ads so we can be annoying, but not too much, Uber system inspiration of course so we don't have to invent entire new architectures, THIS coupon only applies for LONG trips.**

Who are the users you are designing the solution for?

The primary users of the Mashwary is ;

- Parents, Normal parents even strict parents, allowing full control of whether a child can order at this moment or not, 2FA like but for parents to children
- Teenagers after they run 16 who can finally order/schedule/ask for trips without parents or make it that their parents can still get to know, allowing full or partial privacy and safety for all these kinds of teenagers, parents can still get live location of car but NOT the kid
- Old people who need their kids to help them do stuff, they can send requests for children with voice mails or basic messages to their tech obsessed kids and teenagers or adults who can order the rides for them, allowing full safety and adults to help their grandpas/grandmas

What are their needs and challenges?

The needs and challenges faced by these users include:

- **Pet Owners:**
  - Rapidly spreading information about the lost pet to a wide audience.
  - Efficiently coordinating search efforts with the community.

- Accessing timely updates and potential leads.

- **Community Members:**
  - Easily reporting sightings of lost pets.
  - Quickly connecting with pet owners to reunite lost pets.
- **Animal Shelters:**
  - Collaborating with pet owners and the community to increase the chances of finding lost pets.
  - Leveraging technology to streamline their operations and improve efficiency.

Empathy Map

Section II: Ideas and solutions

2.1 Define

What is the final problem statement for your project?
Summarize the key problem that you will focus on solving.

What impact will solving this problem have on users and the market? *Describe how your solution will make life better for users and help the job market or industry*

2.2 Task Review:

In this chapter, we will discuss problem definition, Prior solutions and our selected solution

What are some possible solutions to the problem?
List different ideas your team has come up with to solve the problem.

Have any similar solutions been tried before?
Research and explain other solutions that have addressed the same problem.

Why is your solution the best option?
Explain why your solution is better or more effective than others.

What will your prototype look like?
Draw or describe the design of your prototype.

How will you test your prototype?
Describe the steps you will take to test whether your prototype works as expected.

What materials do you need to build your prototype?
List the materials and their costs for building your prototype.

2.3 Problem Statement:

The core problem we aim to solve is the inefficiency and stress associated with finding lost pets. Currently, individuals often rely on traditional methods like posting flyers and social media, which can be time-consuming, ineffective, and limited in reach.



What impact will solving this problem have on users and the market?

By providing a centralized, efficient, and user-friendly platform for reporting lost pets and connecting with potential finders, we expect the following impacts:

- **Improved User Experience:**
  - Reduced stress and anxiety for pet owners during a difficult time.
  - Increased chances of reuniting lost pets with their owners.
  - Simplified the process of reporting lost pets and helping in the search.
- **Market and Industry Impact:**
  - Potential for partnerships with animal shelters and veterinary clinics to expand the app's reach and services.
  - Opportunities for monetization through premium features or advertising. ○ Positive impact on the animal welfare community by promoting responsible pet ownership and encouraging community involvement.

How will your solution make life better for users and help the job market or industry?

Pet Finder will make life better for users by:

- **Saving time and effort:** The app will streamline the process of reporting lost pets and searching for missing animals.
- **Providing peace of mind:** The app will offer a sense of hope and reassurance to pet owners during a stressful time.
- **Strengthening community bonds:** The app will foster a sense of community and encourage people to help each other.

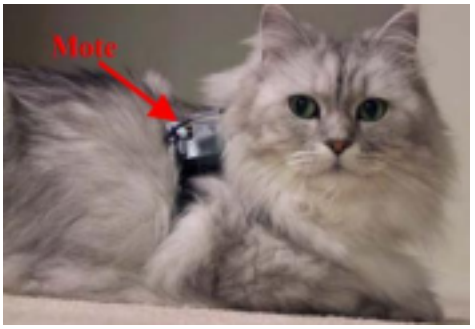
For the job market and industry, Pet Finder has the potential to:

- **Create new job opportunities:** The development and maintenance of the app will require various technical and non-technical skills.
- **Stimulate innovation:** The app can serve as a platform for further innovation in pet tracking and recovery technologies.
- **Contribute to the growth of the pet care industry:** By increasing awareness of pet loss prevention and recovery, the app can contribute to the overall growth of the pet care industry.

2.4 Similar Solutions:

▪ Solution 1: Pet Tracker System

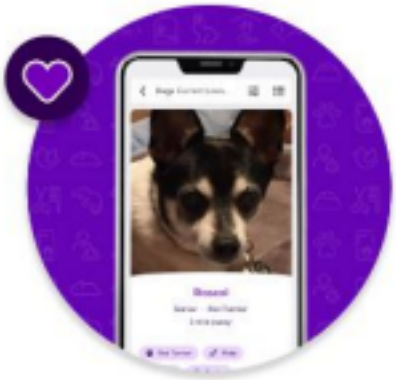
A networked location aware pet tracking and logging application implemented using motes and sensors. (1)



	Advantages Disadvantages
Very precise and accurate	High Cost Hardware Requirements

- **Solution 2: Pets Finder Application**

A Social Media App Help Pet owner to publish Lost Pet Information and Communicate to Pet Finder.



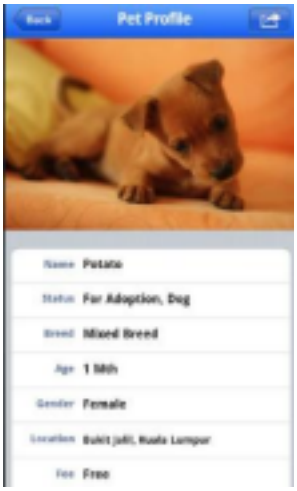
	Advantages Disadvantages
Low Cost	Require more time for finding pets

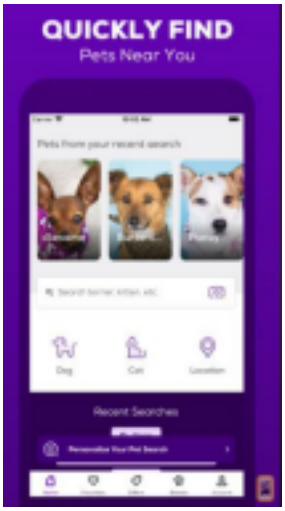
2.5 Ideate:

2.6 Problem Solution:

The solution is nice and easy-to-use product both for the owners and for those who help with finding pets, as they are an important link in a chain of a successful

reunion.





2.7 Our Selected Solution:

A Social Media App Help Pet owner to publish Lost Pet Information and Communicate to Pet Finder.

Creative additions:

- Adaption for pets,
- Advices foe how to keep your pet safe.

2.8 Solution Advantages & Disadvantages:

	Advantages Disadvantages
Low Cost	Require more time for finding pets

2.9 Prototype Design:

Our prototype will be a mobile app with the following features:

- **User Profile:** Users can create profiles to share information about themselves and

their pets.
- **Lost Pet Report:** A form to report a lost pet, including details like the pet's breed, color, and last seen location.
- **Community Feed:** A feed where users can share posts about lost and found pets, as well as safety tips.
- **Notification System:** Real-time notifications for users about relevant updates and alerts.
- **Map Integration:** A map to visualize the location of lost pets and potential sightings.

2.10 Prototype Testing:

To test our prototype, we will conduct user testing with a diverse group of individuals, including pet owners, community members, and animal shelters. We will evaluate the following:

- **Usability:** Is the app easy to navigate and use?
- **Functionality:** Do all features work as intended?
- **Effectiveness:** Does the app help users find lost pets efficiently?
- **User Satisfaction:** Are users satisfied with the app's overall experience?

2.11 Materials and Costs:

To build our prototype, we will need:

- **Development Tools:**
  - IDE (e.g., Android Studio, Xcode): Free

  - Programming Languages (e.g., Java, Kotlin, Swift): Free
- **Design Tools:**
  - Figma, Adobe XD: Free or subscription-based
- **Server Hosting:**
  - Cloud-based services (e.g., AWS, Google Cloud): Varying costs based on usage
- **Testing Devices:**
  - Smartphones and tablets: Varies based on device and quantity

While the initial development costs may vary, the ongoing maintenance and hosting costs can be relatively low, especially if we leverage free or low-cost cloud-based solutions.

2.12 Preliminary Feedback

- **What feedback did you receive from the panel?**
  List feedback you received from the panel about your project.
- **What changes did you make based on the panel's feedback?**
  List any improvements you made to your project after getting feedback.

Section III: Experiment ideas and solutions

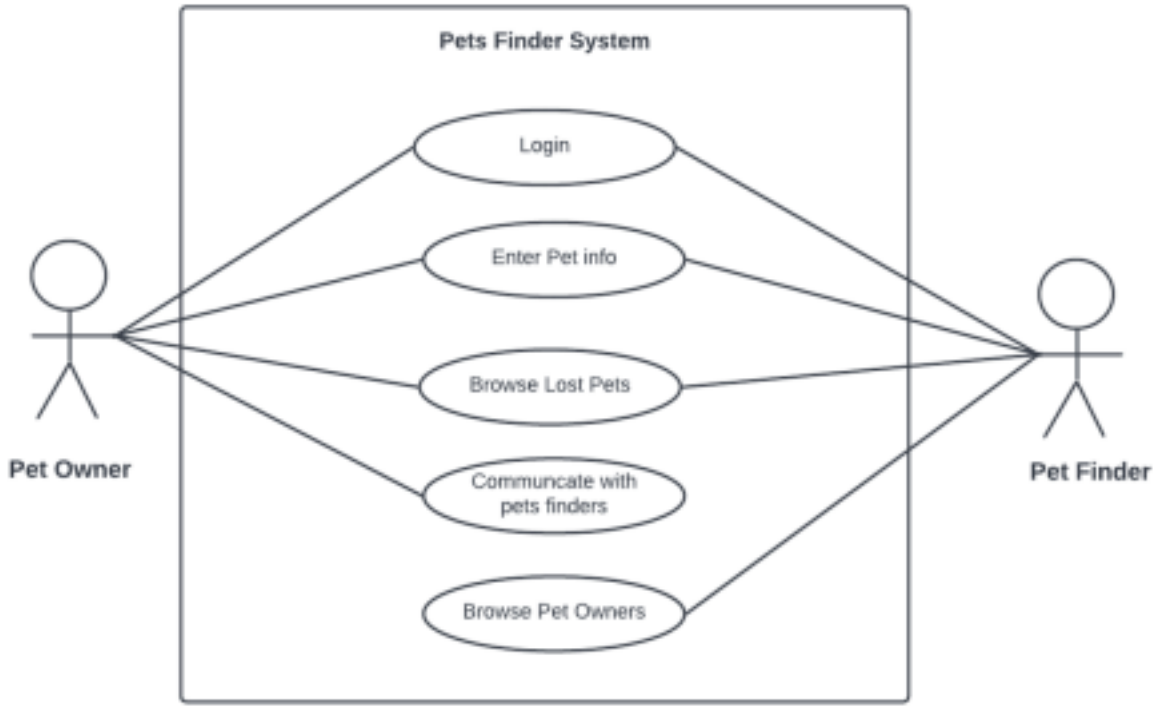
3.1 Task Review:

In this chapter, we will discuss initial system requirements using Use Case and Activity Diagrams. Also describe each Use Case of the system.

3.2 Methods used in Requirement Gathering:

- Interviews or Survey: 50 person of pet owners
- Brainstorming:
- User Observation:

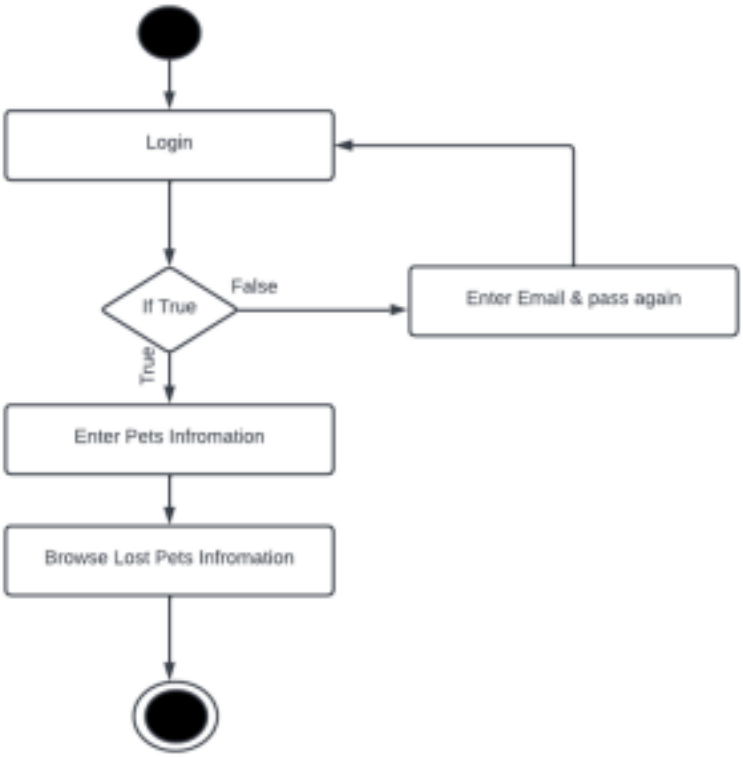
3.3 Use Case Diagram:



Use Case Description:

3.4 Activity Diagram:

First Case: Pet Owner



Activity Diagram Description:

- User login to the system, IF login success:
- User logged-in the system.
- IF not, system ask to register.
- If user logged-in, user can enter pet's information and browse lost pet's information.

3.5 Sketches:

Design the Prototype

● **Draw or describe the final design of your prototype with dimensions.** *Draw or describe the final version of the design for your prototype.* ● **What is the final materials list needed to build your prototype?** *Write the final material list and their costs for building your prototype.* ● **What challenges do you expect during testing?**

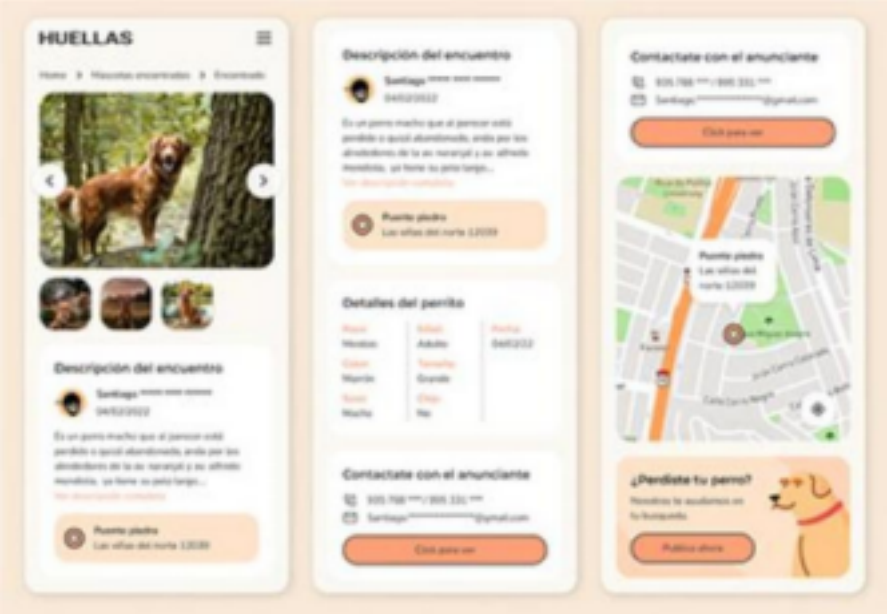
Write down any challenges you think you might face when testing your prototype.

3.6 Linkable Prototype

Prototype Link :

Registration Screen: Where the user will enter his personal information needed to run the application in the way that is designed for. 1. His name

2. Email Address
3. Phone Number
4. Country
5. Location



Building the Prototype

● **What challenges did you face during building the prototype? How did you overcome them?**

Write down any challenges you faced during building your prototype.

3.7 Test and Evaluate:

Use Cases: For example (but not limited): (1)

1. Check the Login functionality
2. Check the Registration functionality
3. Check the Payment functionality
4. Check the Search functionality 5. Check the Shopping functionality

Test Cases:

Test Case ID	Test Case Description		Test Steps Test Data Expected	Results	Actual Results	Pass/Fail
TC01					As or Not as Expected	Pass or Not Pass
TC02					As or Not as Expected	Pass or Not Pass

- **What tests did you run on your prototype?**
  List the steps you followed during testing.

- **What feedback did you receive from testing?**

Summarize what you learned from testing and how users responded.

3.8 Survey and Evaluation:

- **Who is your target audience to fill the survey?**

- **Number of survey respondents?**

Software application Survey:

	Very Satisfactied	Somewhat Satisfied	Neutral	Somewhat Unsatisfied	Very Unsatisfied
Application software compatibility	○	○	○	○	○
Documentation	○	○	○	○	○
Collaborate with team	○	○	○	○	○
Clarity of documentation	○	○	○	○	○
Accessibility of product support	○	○	○	○	○
Quality of product support	○	○	○	○	○
Value for money	○	○	○	○	○
	Very Satisfactied	Somewhat Satisfied	Neutral	Somewhat Unsatisfied	Very Unsatisfied
Overall reliability	○	○	○	○	○
Overall performance	○	○	○	○	○

21

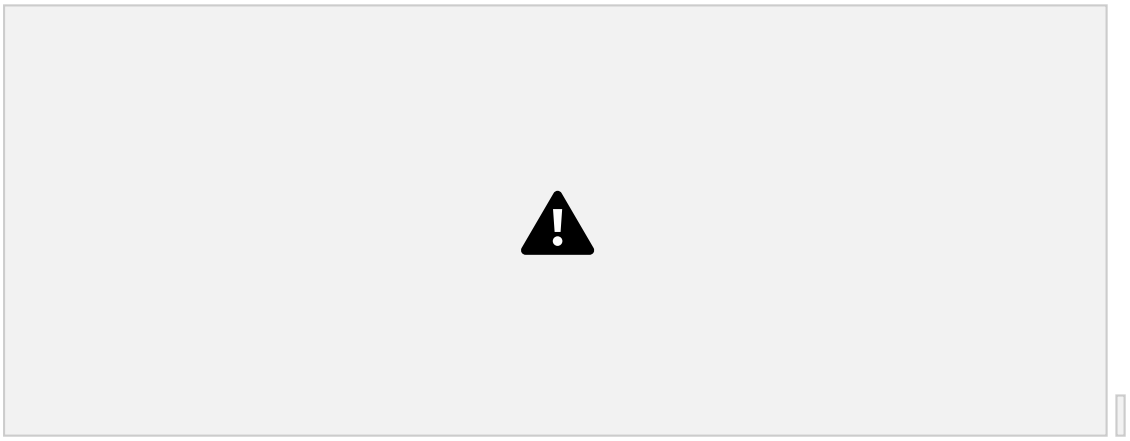


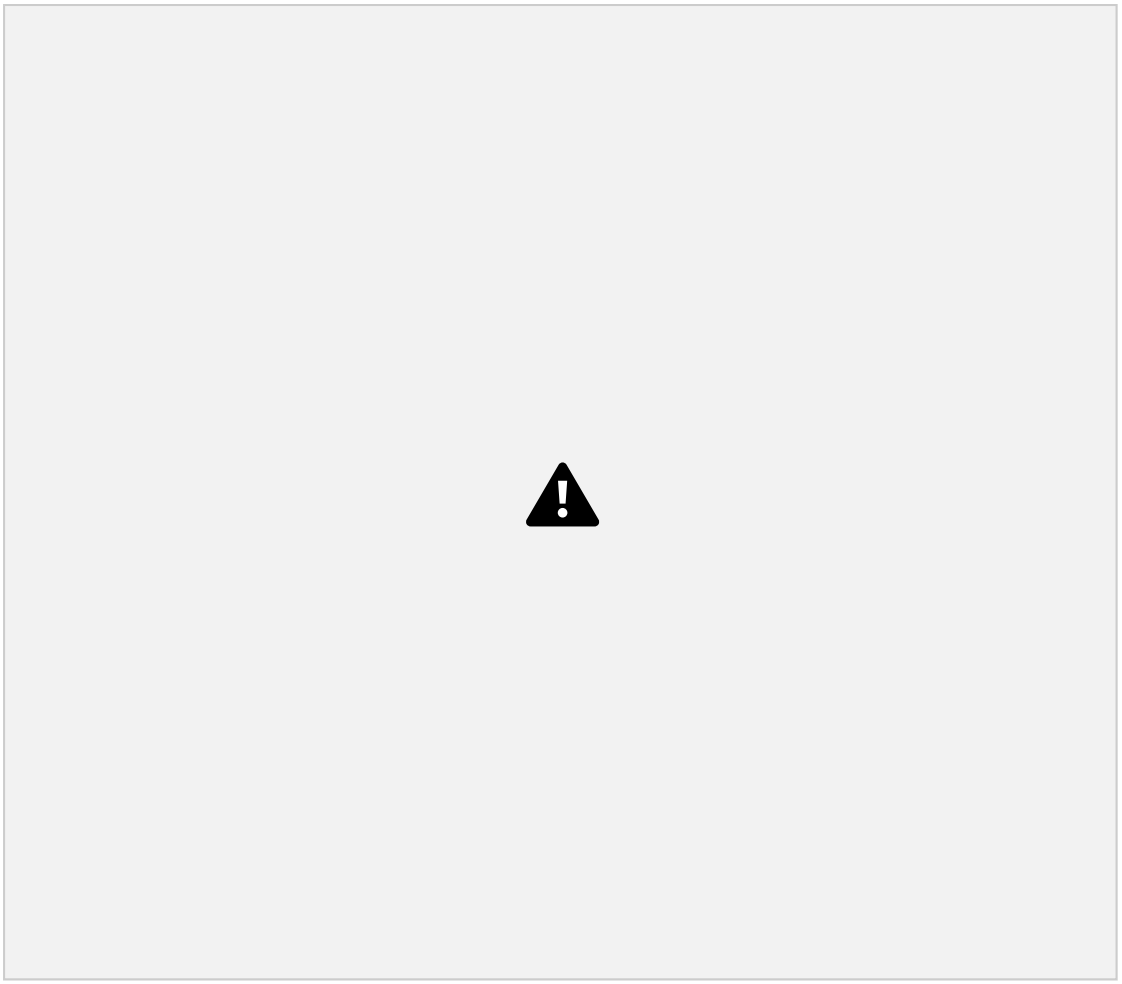
3.8Refine the design.

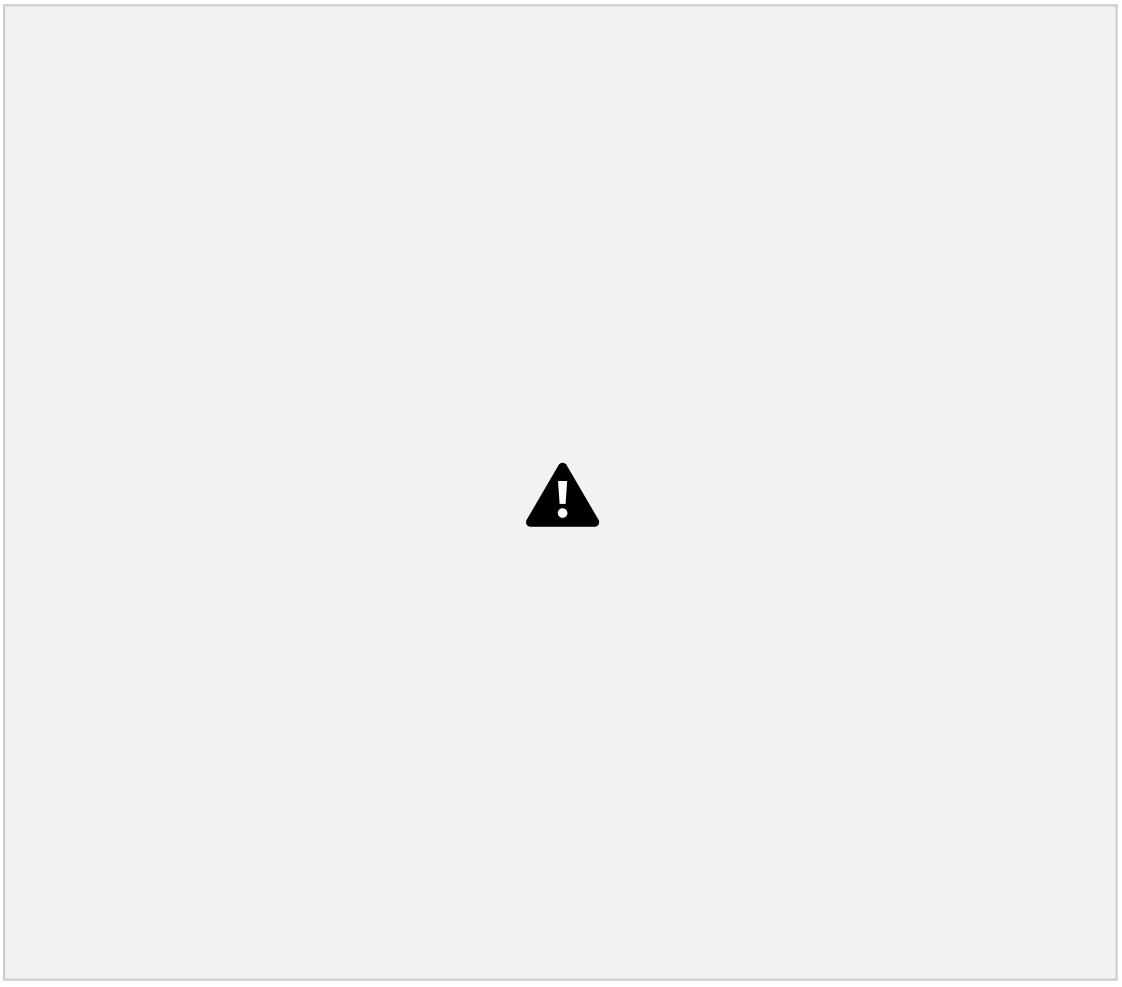
Provide a visual representation of your data and results.
Include a chart, graph, or any visual to display your results.

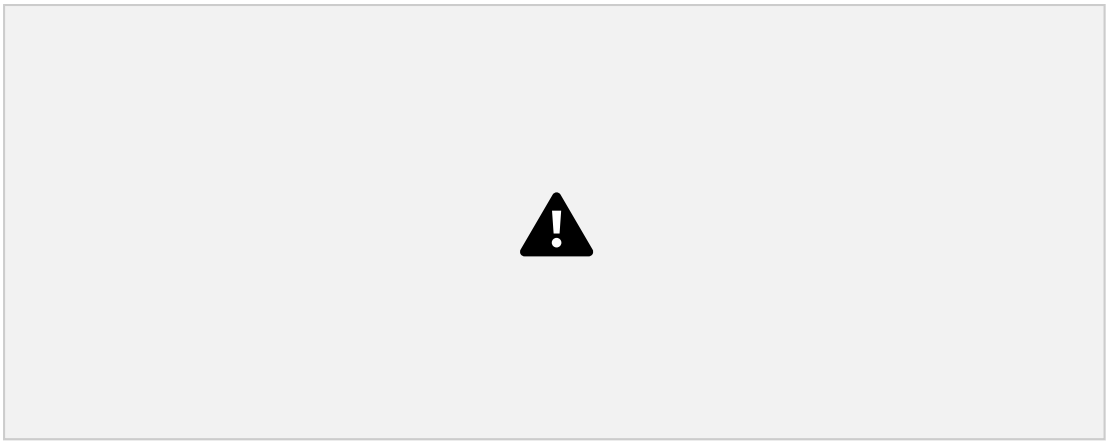
How did you improve your prototype based on testing?
Write down changes you have made to improve your design.

Survey Results:









Section IV: Analyzing and communicating.

4.1 Recommendations

What recommendations do you have for future work in this area?
This includes recommendations on the prototype itself and/or the real life

application of the project idea.

What would you tell another team who wanted to start where you stopped on your solution to help them?

4.2 Learning Outcomes

Transferred academic, Technical, and key Competences:

- **Which competences did you use as related to your Capstone? Explain how each competence was used in your Capstone?**
  Write a brief summary of how did you use the knowledge and skills learned in different competencies in the capstone project.

Competency Related:

▪ Wireframe:

1. Study Use Case Diagram
2. Study Activity Diagram
3. Study Requirements Gathering

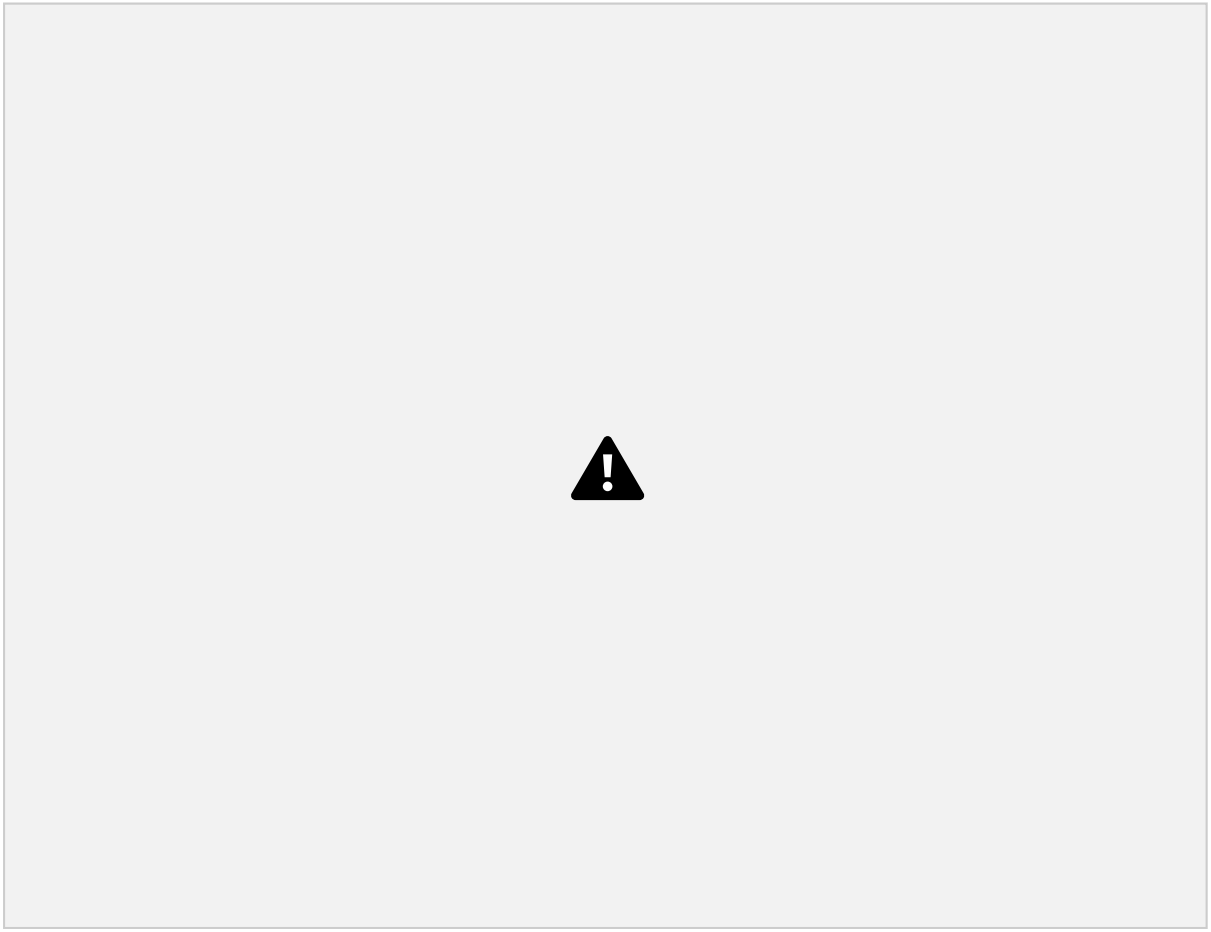
• Testing:

1. Test Scenarios
2. Test Cases

Story to communicate:

- **What is the story behind your project?**
  Write a brief summary of your project's journey, from problem to solution.
- **What did your team learn from this project?**
  Reflect on the skills you gained and the lessons learned throughout the project.
- **How did your project impact your users and the market?**
  Write about the positive effects your project had on users and the broader market.

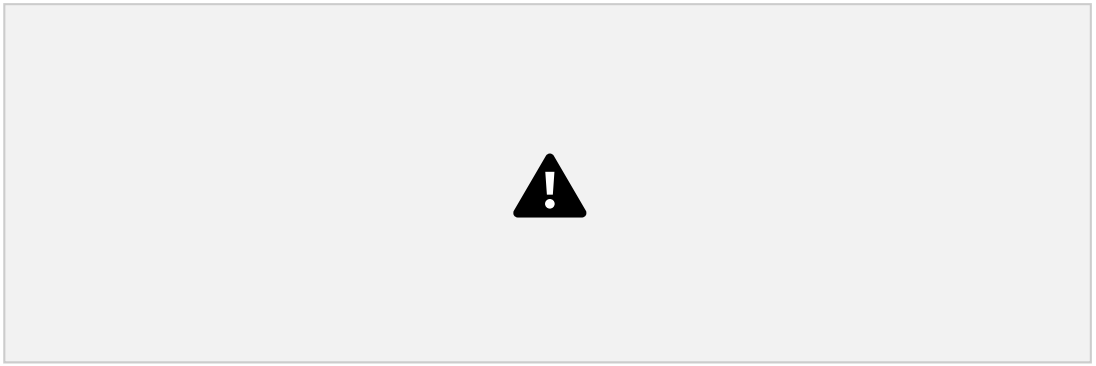
4.3 Communication



26

4.4 References

- **List your sources in APA format as follows:**



References:

(1) BRANDONGAILLE, 29 May 2019, 25 curious homeless pets' statistics,

URL: https://brandongaille.com/24-curious-homeless-pet-statistics/

(2) Zhengming Tang, H. H. (2005). *Pet Tracking System Using Motes*. USA: University of Washington.

(3) Pet finders, founded in 1996, pet finder, URL:https://www.petfinder.com/about/

(4) BRANDONGAILLE, 29 May 2019, 25 curious homeless pets' statistics, URL: https://brandongaille.com/24-curious-homeless-pet-statistics/

Section V: Business Model Canvas Template

